Theory Questions :-

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Q1. What is recursion? How does it work?
Q2. What are base cases and recursive cases in recursion?
Q3. How does recursion work internally in memory (call stack)?
Q4. What is the difference between recursion and iteration?
Q5. What are the advantages and disadvantages of recursion?
Q6. What is the difference between tail recursion and non-tail recursion?
Q7. What is stack overflow in recursion and why does it occur?
Q8. Can you explain the recursion tree concept?
Q9. What is the difference between backtracking and recursion?
Q10. What is memorization? How does it optimize recursive solutions?
Q11. What is Dynamic Programming?
Q12. When should you use Dynamic Programming?
Q13. What are the key characteristics of DP problems?
Q14. What is the difference between recursion and dynamic programming?
Q15. How is DP different from greedy algorithms?
Q16. What are overlapping subproblems?
Q17. What is optimal substructure?
Q18. Why are these properties important for DP?

MCQ Questions :-

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Q1. What will be the output of the following program?

```
int arr[] = {10,20,30,40};
System.out.println(arr[2]);
```

A) 10
B) 20
C) 30
D) 40

Q2. What will be the output?

```
int arr[] = {5,10,15};
for(int i=0;i<arr.length;i++)
{
   System.out.print(arr[i] + " ");
}
```

A) 5 10 15
B) 0 1 2
C) 15 10 5
D) Error

Q3. Which is the correct way to declare an integer array in Java?
A) int arr = new int[5];
B) int arr[] = new int[5];
C) array int arr[5];
D) int[5] arr;
Q4. What will be the output?

```
int arr[] = {2,4,6,8};
```

System.out.println(arr.length);
A) 3
B) 5
C) 4
D) Error

Q5. What happens if we access invalid index?
int arr[] = {1,2,3};
System.out.println(arr[5]);
A) 0
B) Garbage Value
C) Runtime Error
D) Compilation Error

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1. The body weight of seven students of a class is recorded as 54 kg, 78 kg, 43 kg, 82 kg, 67 kg, 42 kg and 75 kg. What is the average body weight of all the seven students?
(a) 63 kg (b) 69 kg (c) 71 kg (d) 73 kg (e) None of these

2. Find the average of the following sets of scores: 385, 441, 876, 221, 536, 46, 291, 428
(a) 221 (b) 403 (c) 428 (d) 536 (e) None of these

3. The monthly incomes of five persons are Rs 1132, Rs 1140, Rs 1144, Rs 1136 and Rs 1148 respectively. What is their arithmetic mean?
(a) Rs 1100 (b) Rs 1120 (c) Rs 1132 (d) Rs 1140

4. Among five people with monthly income (in Rs) 15000, 26000, 16000, 19000 and 50000, how many will have income less than the mean income of five people?
(a) 1 (b) 2 (c) 3 (d) 4

5. The arithmetic mean of 15 numbers is 41.4. Then the sum of these numbers is
(a) 414 (b) 420 (c) 620 (d) 621

6. If 25a + 25b = 115, what is the average of a and b?
(a) 2.5 (b) 3.4 (c) 4.5 (d) 4.6 (e) None of these

7. The following table shows the number of working hours and the number of employees employed in a small scale industry.

No. of working hours	No. of employees
3 – 5	7
5 – 7	10
7 – 9	18
9 – 11	57
11 – 13	14
13 – 15	8

The average number of working hours of an employee is
(a) 8.5 (b) 9.5 (c) 10.5 (d) None of these

8. If the arithmetic mean of 0, 5, 4, 3 is a, that of – 1, 0, 1, 5, 4, 3 is b and that of 5, 4, 3 is c, then the relation between a, b and c is
(a) a = b = c (b) a:b c = 3:2:4 (c) 4a = 5b = c (d) a + b + c = 12

9. Kunal bought 65 books for Rs 1050 from one shop and 50 books for Rs 1020 from another. What

is the average price he paid per book?
(a) Rs 18 (b) Rs 18.20 (c) Rs 24 (d) Rs 36.40 (e) None of these

10. The average of the reciprocals of x and y is
(a) (x+y)/(x-y) (b) (x+y)/2xy (c) 2(x+y)/xy (d) 2xy/(x+y)

Practice Questions :-

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Q1.Write a Java program to calculate the average of all elements present in an integer array.
Explanation

- The **average** of array elements is calculated by:
  Average=Sum of all elementsNumber of elements\text{Average} = \frac{\text{Sum of all elements}}{\text{Number of elements}}Average=Number of elementsSum of all elements
- First, iterate through the array and add all elements to a variable sum.
- Then divide sum by the total number of elements (array.length) to get the average.

Input Array:
[10, 20, 30, 40, 50]
Output:
Average of array elements = 30

Q2. Write a Java program to count the number of even and odd elements present in a given integer array.
Explanation

- An even number is a number that is completely divisible by 2.
- An odd number is a number that is not divisible by 2.
- Traverse the array using a loop.

Input :- Array = { 10, 15, 20, 25, 30 }
Output :- Even count = 3
Odd count = 2

Q3. Write a Java program to check whether a given array is empty or not.

Explanation

1. Every array has a predefined property called length.
2. If array.length == 0, then the array is empty.
3. Otherwise, the array contains elements.

Input :- Array elements: { }
Output :- Array is empty

Q4. Given an integer array, replace all the negative numbers in the array with 0 and print the updated array.

Explanation

- Traverse the array from the first element to the last.
- Check each element:
  - If the element is negative, replace it with 0.
  - If the element is zero or positive, keep it as it is.
- After completing the traversal, print the modified array.

Input :- Array = [5, -3, 7, -1, 0, -6, 4]

Output :- Updated Array = [5, 0, 7, 0, 0, 0, 4]

Q5. Write a Java program to print all elements from an integer array that are greater than a given number.

Explanation

- An integer array is given.
- A number N is also given.
- Traverse the array using a loop.
- Compare each element with N.
- If the element is greater than N, print it.

Input :- Array: 10 25 5 40 18
Given Number: 20

Output :- Elements greater than 20 :
25 40